

EMS4485

USB Type-C Analog Audio Switch with Protection Function, Moisture and Audio Jack Detection

1. General Description

The EMS4485 is a high-performance USB Type-C analog switch for the use in portable multimedia devices, supporting analog audio headsets. The EMS4485 can detect OMTP, CTIA, or 3-pole headsets and configure pinouts automatically. The EMS4485 shares common Type-C pins to pass USB 2.0 signals and analog audio signals; the sideband uses wires and analog microphone signals. The EMS4485 also supports high voltage and surge on SBUx pins and USB pins on the USB Type-C receptacle side. The EMS4485 supports the moisture detection for five pins, including CC_IN, DP_R, DN_L, SBU1 and SBU2. Moisture detection function can detect both moisture resistance and voltage. The detection mode includes single-pin mode and multi-pin mode. The multi-pin mode supports the detection polling, and the polling time can be configured through I2C.

2. Features and Benefits

- USB 2.0 switch -3dB bandwidth: -3dB @950 MHz
- Hi-Fi Audio Path: THD+N= -105dB, 1Vrms, f=20~20kHz, 32Ω Load
- Audio Path Insertion loss: -3dB @900 MHz
- Overvoltage protection: DP_R, DN_L, SBUx and GSBUX, Vth = 4.4 V (default), 4.4 V ~ 5.0 V configurable
- Support OMTP, CTIA and 3-pole Audio Jack Pinout
- Built-in moisture detection function, including moisture resistor and moisture voltage mode. Support single-pin mode and multi-pin mode. Support polling mode, and the polling time can be configured through I2C.

3. Applications

- USB Type-C Receptacle, Mobile Phone and Tablets

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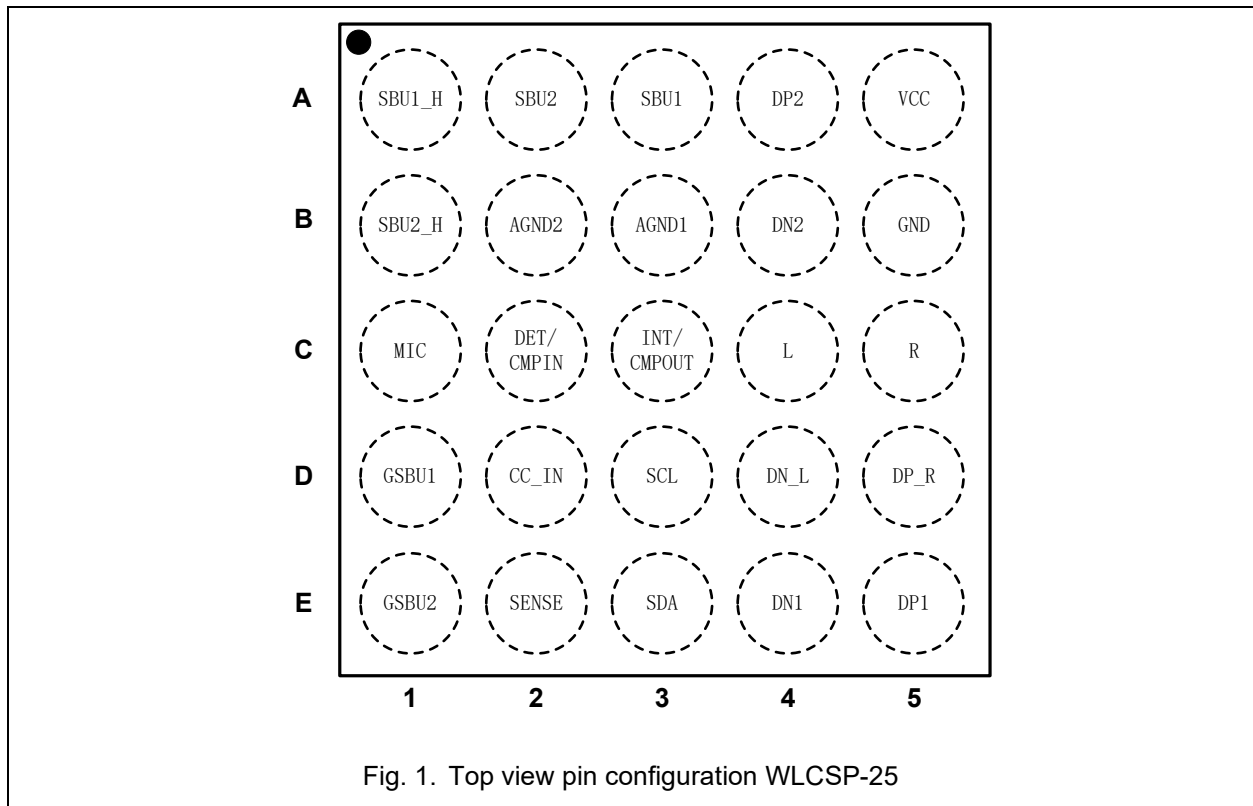
4. Ordering Information

Table 1. Ordering information

Type number	Topside marking	Package		
		Name	Description	Quantity
EMS4485	A6490	WLCSP-25	Wafer-Level Chip Scale Package, 25pins; body width 2 × 2 mm; 0.4 mm pitch, 0.586 mm height	3000

5. Pinning Information

5.1. Pinning



- (1) B2(AGND2), B3(AGND1), B5(GND) pins must be connected together.
- (2) A3(SBU1), D1(GSBU1) pins must be connected together.
- (3) A2(SBU2), E1(GSBU2) pins must be connected together.

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5.2. Pin description

Table 2. Pin description

Pin No.	Name	Type	Description
A5	VCC	PWR	Positive Supply
B5	GND	GND	Primary Ground Connection. Must be Connected to System Ground
D5	DP_R	I/O	Right Audio / Positive USB2.0 Data Common Line
D4	DN_L	I/O	Left Audio / Negative USB2.0 Data Common Line
E5	DP1	I/O	Positive USB2.0 Data Line 1
E4	DN1	I/O	Negative USB2.0 Data Line 1
C5	R	I/O	Right Line for Audio Signals
C4	L	I/O	Left Line for Audio Signals
A3	SBU1	I/O	Sideband Use wire 1
A2	SBU2	I/O	Sideband Use wire 2
C1	MIC	O	Analog Audio Microphone
B2	AGND2	GND	Analog Ground 2
B3	AGND1	GND	Analog Ground 1
E2	SENSE	O	Analog Ground Sense Return
C3	INT/ CMPOUT	O	Open-drain comparator output, set as CMPOUT when 0x12h bit [4] = 1
			Open-drain output, set as INT when 0x12h bit [4] = 0. I2C interrupt output, active low
D2	CC_IN	I	Audio accessory attachment detection input
D1	GSBU1	I/O	Star-connection with SBU1 to Headset Jack as Audio Ground Sense Path 1
E1	GSBU2	I/O	Star-connection with SBU2 to Headset Jack as Audio Ground Sense Path 2
C2	DET/ CMPIN	O	Comparator input, set as COMPIN when 0x12h bit [4] = 1
			Open-drain output, set as DET when 0x12h bit [4] = 0. When CC_IN>1.5v, DET is low. When CC_IN<1.2v, DET is high.
D3	SCL	I	I2C Clock wire
E3	SDA	I/O	I2C Data wire
B1	SBU2_H	I	System Side Sideband Use wire 2, can be configured as eDP AUX path
A1	SBU1_H	I	System Side Sideband Use wire 1, can be configured as eDP AUX path
A4	DP2	I/O	Positive USB2.0 Data Line 2
B4	DN2	I/O	Negative USB2.0 Data Line 2

- (1) B2(AGND2), B3(AGND1), B5(GND) pins must be connected together.
- (2) A3(SBU1), D1(GSBU1) pins must be connected together.
- (3) A2(SBU2), E1(GSBU2) pins must be connected together.

6. Functional Diagram

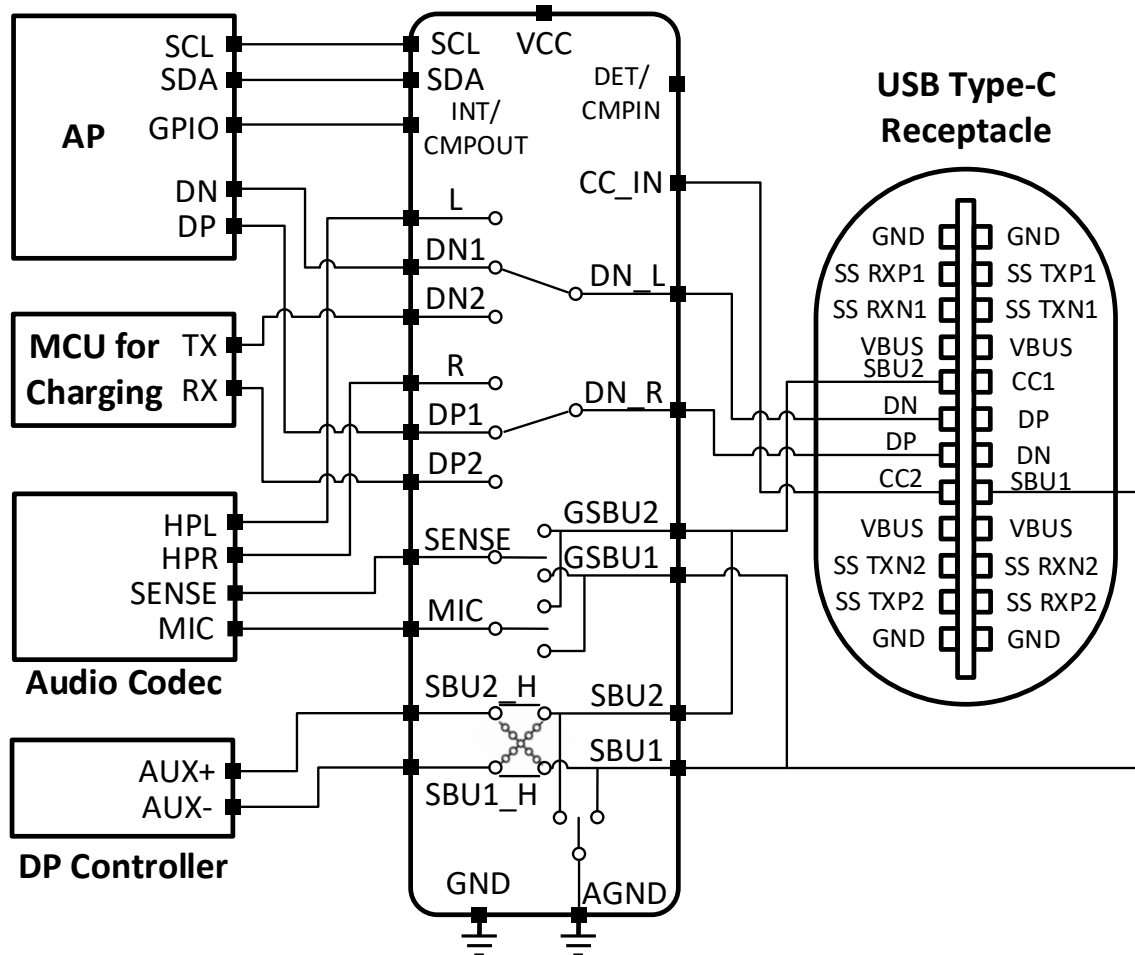


Fig. 2. Functional Diagram

- (1) B2(AGND2), B3(AGND1), B5(GND) pins must be connected together.
- (2) A3(SBU1), D1(GSBUS1) pins must be connected together.
- (3) A2(SBU2), E1(GSBUS2) pins must be connected together.

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7. Absolute Maximum Ratings

Stresses beyond those listed in Table-3 Absolute Maximum Ratings may cause permanent damage to the device. They are stress ratings only, which do not imply functional operation of the device at these or any other conditions. Beyond those indicated under Recommended Operating Conditions, exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.

Table 3. Absolute Maximum Ratings

Over operating free-air temperature range (unless otherwise noted).

Symbol	Parameter		Min	Max	Unit
V _{CC}	Supply voltage from V _{CC}		-0.5	5.5	V
V _{CC_IN}	V _{CC_IN} , to GND		-0.5	20	V
V _{SW_C}	V _{DP_R} to GND, V _{DN_L} to GND		-3.3	16	V
V _{SW_Audio}	V _L to GND, V _R to GND		-3.3	5.5	V
V _{SW_USB}	V _{DP1} to GND, V _{DN1} to GND		-0.3	5.5	V
V _{SW_UART}	V _{DP2} to GND, V _{DN2} to GND		-0.3	5.5	V
V _{V_SBUx/GSBUx}	V _{SBU1} to GND, V _{SBU2} to GND, V _{GSBU1} to GND, V _{GSBU2} to GND		-0.3	16	V
V _{VSBUx_H}	V _{SBU1_H} to GND, V _{SBU2_H} to GND		-0.3	5.5	V
V _{I/O}	SENSE, MIC to GND		-0.3	5.5	V
V _{CNTRL}	Control input voltage	SDA, SCL	-0.5	5.5	V
V _{comparator}	Comparator input and output	DET/CMPIN, INT/CMPOUT	-0.5	5.5	V
I _{SW_Audio}	Switch I/O current, audio path		-250	250	mA
I _{SW_USB}	Switch I/O current, USB path			100	mA
I _{SW_UART}	Switch I/O current, UART path			50	mA
I _{SW_MIC}	Switch I/O current, MIC to GSBU1 or GSBU2			50	mA
I _{SW_SBUx}	Switch I/O current, SBUx to SBUx_H			50	mA
I _{SW_SENSE}	Switch I/O current, SENSE to GSBU1 or GSBU2			100	mA
I _{SW_AGND}	Switch I/O current, AGND to SBU1 or SBU2			500	mA
T _A	Absolute maximum operating temperature		-40	85	°C
T _{STG}	Storage temperature		-55	150	°C

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8. Recommended Operating Conditions

Table 4. Recommended Operating Conditions

Symbol	Parameter	Min	Max	Unit
V _{CC}	Supply voltage	3	4.5	V
V _{SW_USB}	V _{DP1} to GND, V _{DN1} to GND, V _{DP_R} to GND, V _{DN_L} to GND	0	5	V
V _{SW_UART}	V _{DP2} to GND, V _{DN2} to GND, V _{DP_R} to GND, V _{DN_L} to GND	0	V _{CC} +0.3	V
V _{SW_Audio}	V _{DP_R} to GND, V _{DN_L} to GND, V _L to GND, V _R to GND	-3.3	5	V
V _{VSBU}	V _{SBU1} to GND, V _{SBU2} to GND	0	V _{CC} +0.3 (3.6V Max)	V
V _{VGSBU}	V _{G SBU1} to GND, V _{G SBU2} to GND	0	3.6	V
V _{SEN_MIC}	V _{SENSE} to GND, V _{MIC} to GND	0	5	V
V _{VSBU_H}	V _{SBU1_H} to GND, V _{SBU2_H} to GND	0	5	V
V _{CC_IN}	V _{CC_IN} to GND	0	18	V
V _{IH}	Input voltage high	1.3	V _{CC} +0.3	V
V _{IL}	Input voltage low		0.3	V
T _A	Ambient operating temperature	-40	85	°C

1. The input and output negative voltage ratings may be exceeded if the input and output diode current ratings are observed.
2. Control input must be held high or Low, it must not float.
3. In USB mode, any signal applied to the off-state audio inputs R, L may not swing below ground or above 1V.

9. ESD Ratings

When a statically-charged person or object touches an electrostatic discharge sensitive device, the electrostatic charge might be drained through sensitive circuitry in the device. If the electrostatic discharge possesses sufficient energy, damage might occur to the device due to localized overheating.

Table 5. ESD Ratings

Parameter	Conditions	Value	Unit
ESD voltage	HBM, DP_R, DN_L, SENSE, MIC pins	±5	kV
	HBM, all other pins	±2.5	kV
	Latch-up	800	mA

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10. Electrical Characteristics

10.1. DC electrical characteristics

The values are obtained under these conditions unless otherwise specified: $V_{CC} = 2.7\text{ V to }4.5\text{ V}$, $V_{CC} = 3.3\text{ V (Typ.)}$, $T_A = -40^{\circ}\text{C to }85^{\circ}\text{C}$, and $T_A = 25^{\circ}\text{C (Typ.)}$

Symbol	Parameter	Conditions	Power	Min	Typ	Max	Unit
I _{CC}	Supply current	USB switches on, SBUx to SBUx_H switches on	V _{CC} = 4.2 V		55		μA
		Audio switches on, MIC switch on and Audio GND switch on			68		μA
I _{CCZ}	Standby current	04H'b7 = 0			5		μA
USB/Audio common pins: DP_R, DN_L							
I _{OZ}	Off leakage current of DP_R and DN_L	DN_L, DP_R = 0 V to 3.6 V	V _{CC} = 2.7 V to 4.5 V	-3		3	μA
I _{OFF}	Power-off leakage current of DP_R and DN_L	DN_L, DP_R = 0 V to 3.6 V	Power off	-3		3	μA
V _{OV_TRIP}	Input OVP lockout	Rising edge, 0x12h bit [7:6] = 00	V _{CC} = 2.7 V to 4.5 V		4.4		V
V _{OV_HYS}	OVP hysteresis				0.4		V
Audio switch							
I _{ON}	On leakage current of audio switch	DN_L, DP_R = -3 V to 3 V, DP, DN, R, L = Float	V _{CC} = 2.7 V to 4.5 V	-5		5	μA
R _{ON}	Switch on resistance	I _{SW} = 100 mA, V _{SW} = -3 V to 3 V	V _{CC} = 2.7 V to 4.5 V		1		Ω
R _{SHUNT}	Pull-down resistor on R/L pin when audio switch is off	L= R = 3 V			10		kΩ
USB switch							

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I_{ON}	On leakage current of USB switch	DN_L, DP_R = 0 V to 3.6 V, DP1, DN1, R, L = Float	$V_{CC} = 2.7 \text{ V to } 4.5 \text{ V}$	-3		3	μA
I_{OZ}	Off leakage current of DP1 and DN1	DP1, DN1 = 0 V to 3.6 V	$V_{CC} = 2.7 \text{ V to } 4.5 \text{ V}$	-3		3	μA
I_{OFF}	Power-off leakage current of DP1 and DN1	DP1, DN1 = 0 V to 3.6 V	Power off	-3		3	μA
R_{ON_USB}	USB switch on resistance	$I_{SW} = 8 \text{ mA}$, $V_{SW} = 0.4 \text{ V}$	$V_{CC} = 2.7 \text{ V to } 4.5 \text{ V}$		6		Ω
UART switch							
I_{OZ}	Off leakage current of DP2 and DN2	DP2, DN2 = 0 V to 3.6 V	$V_{CC} = 2.7 \text{ V to } 4.5 \text{ V}$	-3		3	μA
R_{ON_UART}	UART switch on resistance	$I_{SW} = 3 \text{ mA}$, $V_{SW} = 0.4 \text{ V}$	$V_{CC} = 2.7 \text{ V to } 4.5 \text{ V}$		17		Ω
I_{ON}	On leakage current UART switch	DN_L, DP_R = 0 V to 3.6 V, DP2, DN2, R, L = Float	$V_{CC} = 2.7 \text{ V to } 4.5 \text{ V}$	-3		3	μA
SENSE switch							
I_{ON}	SENSE path leakage current	GSBUx = 0 V to 1 V, SENSE is floating	$V_{CC} = 2.7 \text{ V to } 4.5 \text{ V}$	-2		2	μA
R_{ON}	SENSE switch on resistance	$I_{OUT} = 100 \text{ mA}$, $V_{SW} = 1.0 \text{ V}$	$V_{CC} = 2.7 \text{ V to } 4.5 \text{ V}$		1.2		Ω
I_{OFF}	Power-off leakage current of SENSE	SENSE = 0 V to 1.0 V	Power off	-2		2	μA
	Power-off leakage current of GSBUX	GSBUx = 0 V to 3.6 V		-3		3	μA
V_{OV_TRIP}	Input OVP lockout on GSBUX	Rising edge, 0x12h bit [7:6] = 00	$V_{CC} = 2.7 \text{ V to } 4.5 \text{ V}$		4.4		V

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V _{OV_HYS}	Input OVP hysteresis of GSBUX				0.4		V
SBUx pins							
I _{OZ}	Off leakage current of SBUx	SBUx = 0 V to 3.6 V	V _{CC} = 2.7 V to 4.5 V	-3		3	μA
I _{OFF}	Power-off leakage current port SBUx	SBUx = 0 V to 3.6 V	Power off	-3		3	μA
V _{OV_TRIP}	Input OVP lockout	Rising edge, 0x12h bit [7:6] = 00	V _{CC} = 2.7 V to 4.5 V		4.4		V
V _{OV_HYS}	Input OVP hysteresis				0.4		V
MIC switch							
I _{ON}	On leakage current of MIC switch	SBUx = 0 V to 3.6 V, MIC is floating	V _{CC} = 2.7 V to 4.5 V	-3		3	μA
I _{OFF}	Power-off leakage current of MIC	MIC = 0 V to 3.6 V	Power off	-1		1	μA
R _{ON}	MIC switch on resistance	V _{SW} =1V, I _{SW} = 30 mA	V _{CC} = 2.7 V to 4.5 V		11		Ω
SBUx_H switch							
I _{ON}	On leakage current of SBUx_H switch	SBUx = 0 V to 3.6 V, SBUx_H is floating	V _{CC} = 2.7 V to 4.5 V	-3		3	μA
I _{OFF}	Power off leakage current of SBUx_H	SBUx_H = 0 V to 3.6 V	Power off	-1		1	μA
R _{ON}	SBUx_H switch on resistance	V _{SW} = 0 V to 3.6 V, I _{SW} = 30 mA	V _{CC} = 2.7 V to 4.5 V		10		Ω
Audio ground switch pin: AGND to SBUx							
R _{ON}	AGND switch on resistance	I _{SOURCE} = 100 mA on SBUx	V _{CC} = 2.7 V to 4.5 V		150		mΩ
CC_IN pin							

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V_{TH_L}	Input low threshold		VCC = 2.7 V to 4.5 V		1.2		V
V_{TH_H}	Input high threshold				1.5		V
I_{IN}	Input leakage of CC_IN	CC_IN = 0 V to 5.5 V		-3		1	μ A
SDA, SCL pins							
V_{ILI2C}	Low-level input voltage		VCC = 2.7 V to 4.5 V			0.3	V
V_{IHI2C}	High-level input voltage			1.3			V
I_{I2C}	Input current of SDA and SCL pins	SCL/SDA = 0 V to 3.6 V		-5		5	μ A
V_{OLSDA}	Low-level output voltage	I_{OL} = 2 mA				0.3	V
I_{OLSDA}	Low-level output current	V_{OLSDA} = 0.2 V		10			mA

Note:

- (4) Flatness is defined as the difference between maximum and minimum value of ON-resistance at the specified analog signal voltage points.
- (5) RON matching between channels is calculated by subtracting the channel with the lowest max Ron value from the channel with the highest max Ron value.
- (6) The OVP voltage can be adjusted by bit [7:6] for 12h: 4.4 V, 4.6 V, 4.8 V and 5 V, each step is 0.2V.
- (7) Specifications subject to change without notice.

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10.2. AC electrical characteristics

The values are obtained under these conditions unless otherwise specified: VCC = 2.7 V to 4.5 V, VCC = 3.3 V (Typ.), TA = -40°C to 85°C, and TA = 25°C (Typ.)

Symbol	Parameter	Conditions	Power	Min	Typ	Max	Unit
Audio switch							
t _{delay}	Audio switch turn-on delay time	DP_R = DN_L = 1 V, R _L = 32 Ω	VCC = 3.3 V		410		μs
t _{rise}	Audio switch turn-on rising time				230		μs
t _{OFF}	Audio switch turn-off time				45		ns
X _{TALK} ⁽¹⁾	Cross talk (adjacent)	f = 1 kHz, R _L = 50 Ω, V _{SW} = 1 V _{RMS}	VCC = 3.3 V		-88		dB
BW	−3 dB bandwidth	R _L = 50 Ω			900		MHz
O _{IRR}	Off isolation	f = 1 kHz, R _L = 50 Ω, C _L = 0 pF, V _{SW} = 1 V _{RMS}			-108		dB
THD+N	Total harmonic distortion + noise performance with A-weighting filter	R _L = 600 Ω, f = 20 Hz ~ 20 kHz, V _{SW} = 2 V _{RMS}			-108		dB
		R _L = 32 Ω, f = 20 Hz ~ 20 kHz, V _{SW} = 1 V _{RMS}			-105		dB
		R _L = 16 Ω, f = 20 Hz ~ 20 kHz, V _{SW} = 0.5 V _{RMS}			-101		dB
USB switch							
t _{ON}	USB switch turn-on time	DP_R = DN_L = 1.5 V, R _L = 50 Ω	VCC = 3.3 V		120		ns
t _{OFF}	USB switch turn-off time				13		ns
BW	−3 dB bandwidth	R _L = 50 Ω			950		MHz
O _{IRR}	Off isolation between DP, DN and common node pins	f = 1 kHz, R _L = 50 Ω, C _L = 0 pF, V _{SW} = 1 V _{RMS}			-107		dB

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t_{OVP}	DP_R and DN_L pins OVP response time	$V_{SW} = 3.5 \text{ V to } 5.5 \text{ V}$			1		μs
UART switch							
t_{ON}	UART switch turn-on time	$DP_R = DN_L = 1.5 \text{ V}, R_L = 50 \Omega$	$VCC = 3.3 \text{ V}$		90		ns
t_{OFF}	UART switch turn-off time				6		ns
BW	-3 dB bandwidth	$R_L = 50 \Omega$			720		MHz
MIC / Audio ground switch							
t_{delay_MIC}	MIC switch turn-on delay time	$SBUx = 1 \text{ V}, R_L = 50 \Omega$	$VCC = 3.3 \text{ V}$		1.75		ms
t_{rise_MIC}	MIC switch turn-on rising time				1.5		ms
t_{OFF_MIC}	MIC switch turn-off time				30		ns
t_{delay_AGND}	AGND switch turn-on time	$SBUx$ pulled to 0.2 V by 8 Ω , AGND connect to GND	$VCC = 3.3 \text{ V}$		430		μs
t_{rise_AGND}	AGND switch turn-on rising time				280		μs
t_{OFF_AGND}	AGND switch turn-off time	$SBUx: I_{SOURCE} = 10 \text{ mA}, \text{ clamp to } 2.5 \text{ V}$			140		ns
BW	MIC switch bandwidth	$R_L = 50 \Omega$			55		MHz
SBUx_H switch							
t_{ON}	SBUx_H switch turn-on time	$SBUx = 2.5 \text{ V}, R_L = 50 \Omega$	$VCC = 3.3 \text{ V}$		320		μs
t_{OFF}	SBUx_H switch turn-off time				12		ns
BW	Bandwidth	$R_L = 50 \Omega$			45		MHz
t_{OVP}	SBUx pins OVP response time	$V_{SW} = 3.5 \text{ V to } 5.5 \text{ V}$	$VCC = 3.3 \text{ V}$		0.8		μs

Note:

(1) Crosstalk is inversely proportional to source impedance.

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10.3. Comparator electrical characteristics

The values are obtained under these conditions unless otherwise specified: VCC = 2.7 V to 4.5 V, VCC = 3.3 V (Typ.) and TA = 25°C (Typ.)

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
Offset voltage						
V _{REF}	Reference voltage	VCC = 2.7 V to 4.5 V		216		mV
V _{HYS}	Hysteresis	VCC = 2.7 V to 4.5 V		20		mV
Input voltage range						
V _{CM}	Common-mode voltage range	VCC = 2.7 V to 4.5 V	-0.1		VCC + 0.1	V

Note:

(1) Guaranteed by design.

10.4. Capacitance

The values are obtained under these conditions unless otherwise specified: VCC = 3.3 V (Typ.) and TA = 25°C (Typ.)

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
CON_USB/Audio	On capacitance (common port)	f = 1 MHz, 100 mV _{PK-PK} , 100 mV DC bias		22		pF
COFF_USB/Audio	Off capacitance (common port)	f = 1 MHz, 100 mV _{PK-PK} , 100 mV DC bias		5.6		pF
COFF_USB	Off capacitance (non-common ports)	f = 1 MHz, 100 mV _{PK-PK} , 100 mV DC bias		0.5		pF
CON_SENSE_SW	On capacitance (common port)	f = 1 MHz, 100 mV _{PK-PK} , 100 mV DC bias		38		pF
CON_MIC_SW	On capacitance (common port)	f = 1 MHz, 100 mV _{PK-PK} , 100 mV DC bias		25		pF
COFF_SENSE /MIC_SW	Off capacitance (common port)	f = 1 MHz, 100 mV _{PK-PK} , 100 mV DC bias		6.8		pF
CON_SBUx_H_SW	On capacitance (common port)	f = 1 MHz, 100 mV _{PK-PK} , 100 mV DC bias		59		pF
COFF_SBUx_H_SW	Off capacitance (common port)	f = 1 MHz, 100 mV _{PK-PK} , 100 mV DC bias		40		pF

Note:

(1) Guaranteed by design.

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10.5. I2C specification

10.5.1. I2C Controlling

EMS4485 switching functions are controlled by 2 I2C pins: SCL and SDA pin. The timing characteristics and diagrams, as well as internal register meaning and usage are listed below:

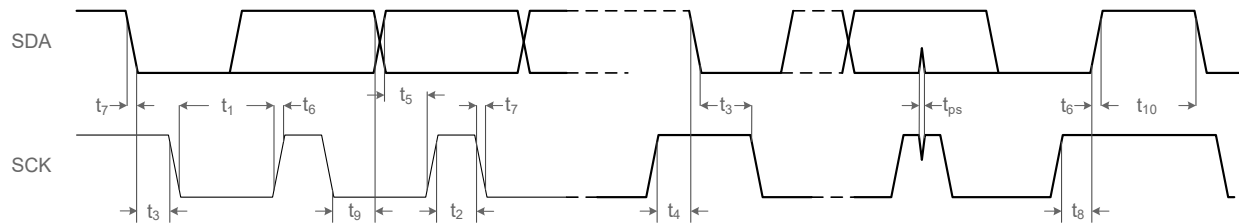


Fig. 3. Definition of timing for full-speed mode devices on the I2C bus

Note: Each of SDA and SCL pins should be pulled up by a 2.2k Ω resistor.

10.5.2. I2C Timing Characteristics

The values are obtained under these conditions unless otherwise specified: VCC = 3.3 V (Typ.) and TA = 25°C (Typ.)

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
f_{CLK}	SCL Frequency				400	kHz
t_1	SCL Low Pulse-Width		1.3			μ s
t_2	SCL High Pulse-Width		0.6			μ s
t_3	Hold Time (Start Condition)		0.6			μ s
t_4	Setup Time (Start Condition)		0.6			μ s
t_5	Data setup time		0.1			μ s
t_6	SDA, SCL Rise Time				0.3	μ s
t_7	SDA, SCL Fall Time				0.3	μ s
t_8	Setup Time (Stop Condition)		0.6			μ s
t_9	Data Hold Time				0.9	μ s
t_{10}	Bus-free time between Stop and Start conditions		1.3			μ s
t_{ps}	Pulse Width of Spikes being Suppressed				50	ns

11. Application Information

11.1. Overvoltage protection

The EMS4485 features an over voltage protection (OVP) on receptacle side pins that turns off the internal signal routing path if the voltage exceeds the OVP threshold. If an OVP is occurred, flag register 0x02h and 0x03h will indicate which pin had the OVP event. OVP threshold voltage is configurable by 0x12h, bit [7:6].

11.2. Watchdog timer

The function of watchdog is to continuously monitor I2C communication. A timer will start as long as the watchdog is enabled. The watchdog receives either read or write command from the master to be reset with I2C communication. In case there's no I2C communication detected after the timer counting down to zero, EMS4485 will control Register 0x04h, 0x05h and 0x12h bit[1] to their default value, reset switch setting and moisture detection enable/disable signals. Watchdog timer can be programmed through Register 0x20h.

11.3. Headset detection

The EMS4485 integrates headset unplug detection function by detecting the CC_IN voltage. The function will be active when device is enabled. Register 0x11h, bit [1:0] Output can indicate if CC_IN is low ($CC_IN < 1.2\text{ V}$) or high ($CC_IN > 1.5\text{ V}$).

	0x11h, bit [1]	0x11h, bit [0]
CC_IN < 1.2 V	0	1
CC_IN > 1.5 V	1	0

11.4. MIC switch auto-off function

The function is active during control bit 0x12h bit [2] = 1. When CC_IN is changed from low to high, and L, R, and AGND switches are on, the MIC switch will be off.

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11.5. Audio jack detection and configuration

The function is active when the headset is inserted and CC_IN is changed from high to low. The EMS4485 can detect OMTP, CTIA or 3-pole headset and configure pinout automatically. During detection and configuration, the R, L, Sense, MIC and Audio ground switches will be off. After detection and configuration, R, L, MIC, Sense and AGND switches will turn on according to detection results and timing control setting.

11.6. I²C interface

The EMS4485 includes a full I²C slave controller. The I²C slave fully complies with the I²C specification version 2.1 requirements. This block is designed for fast mode, 400 kHz, signals. Examples of an I²C write and read sequence are shown in below figures respectively.

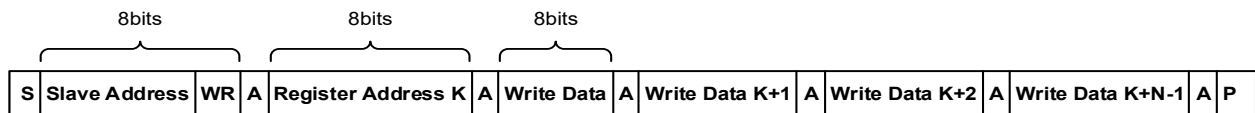


Fig. 4. I²C write example

Note: Single-byte read is initiated by the master with P immediately following the first data byte.

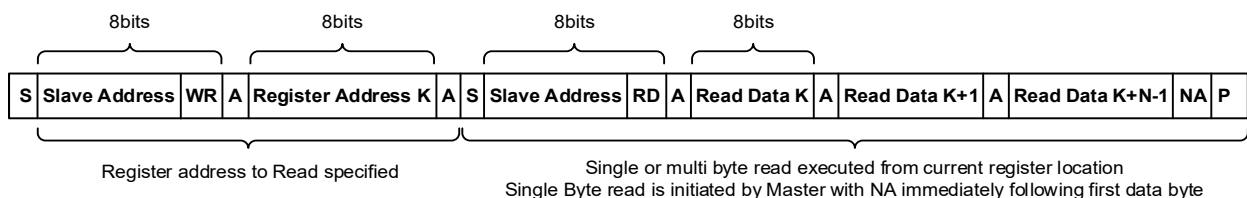


Fig. 5. I²C read example

Note: If the register is not specified, the master will begin reading from the current register. In this case only sequence showing in the red bracket is needed.

11.7. Moisture detection

The EMS4485 can detect moisture in type-C connector. When moisture detection is enabled, all the related channels will be closed. The voltage can be measured from the selected pin with current source enabled or disabled according to 0x13h bit [3]. Leakage between the selected pins and GND/VBUS can be detected by comparing the measured results with pre-set thresholds, and

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it will pull down “INT”, when moisture is found. The related channels will be recovered when the pin detection is completed.

An internal current source will be applied to the pin first during detection of the leakage between GND and pin CC_IN, DP_R, DN_L, SBU1 or SBU2. After getting the result reflecting the resistance from this pin to GND, the result will be stored to corresponding register.

The voltage from the pin without current source will be measured during detection of the leakage between VBUS and pin CC_IN, DP_R, DN_L, SBU1 or SBU2. For the leakage from VBUS and the voltage drop from this pin to ground, they will be detected by the measurement when the moisture detection is performed. The moisture detection has single-pin mode and multi-pin mode.

Single-pin mode

In single-pin mode, the moisture detection result would be stored separately in register 0x14h (with current source enabled) or 0x30h (with current source disabled).

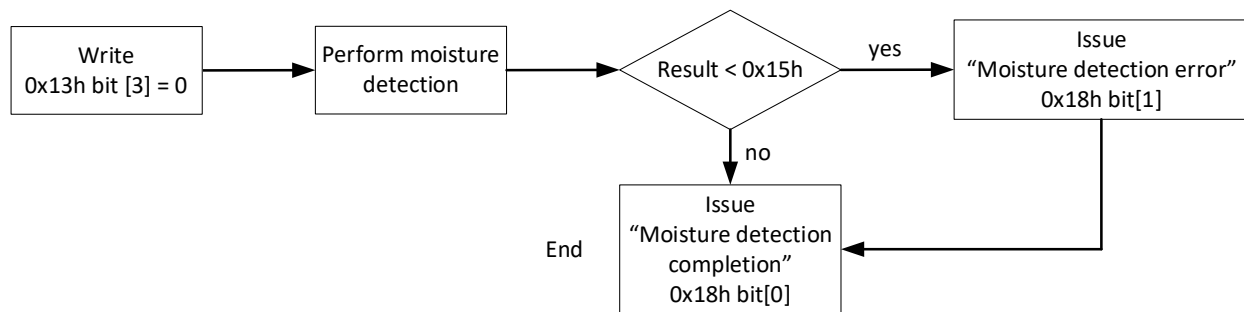


Fig. 6. Single-pin mode (with current source enabled)

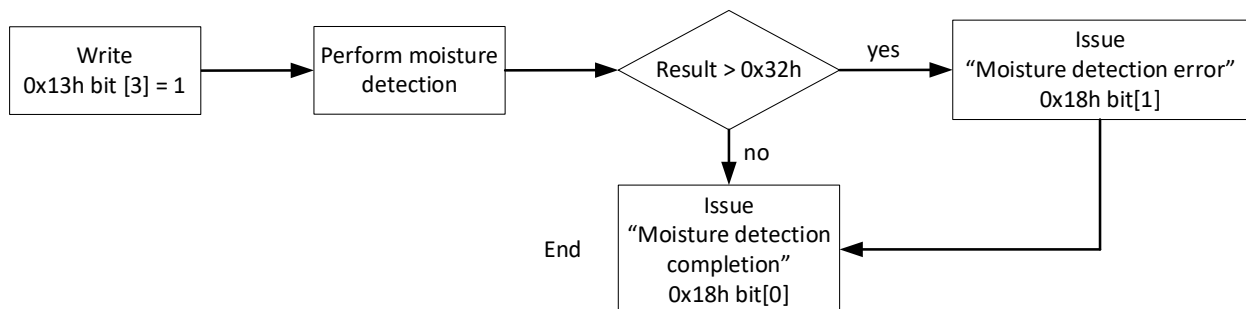


Fig. 7. Single-pin mode (with current source disabled)

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Multi-pin mode

In multi-pin mode, if all pins are selected, the turn of moisture detecting sequence will be CC_IN, DP_R, DN_L, SBU1, SBU2. In case some pins are not selected, the detection will ignore the unselected pins and performed on the next selected pins. For example, if the pin CC_IN, DP_L are not selected, the moisture detection will be performed on DP_R first, then SBU1 and SBU2 in turn.

As soon as the detection starts, whether the setup of the current source or pin selection would be ignored. And all the related channels would be forced off during the whole round detection (from the 1st pin to the last pin). The related channels will be recovered after the detection for all pins are completed.

If moisture error is detected in some pins (result > 0x32h for detection with current source disabled, or the result < 0x15h with current source enabled), EMS4485 will complete the whole round, and then stop moisture detection and indicate the flag of “Moisture detection completion”. 0x31h bit [5] will select the blank time between two adjacent detections. It’s shown in fig 6, fig 7 as below.

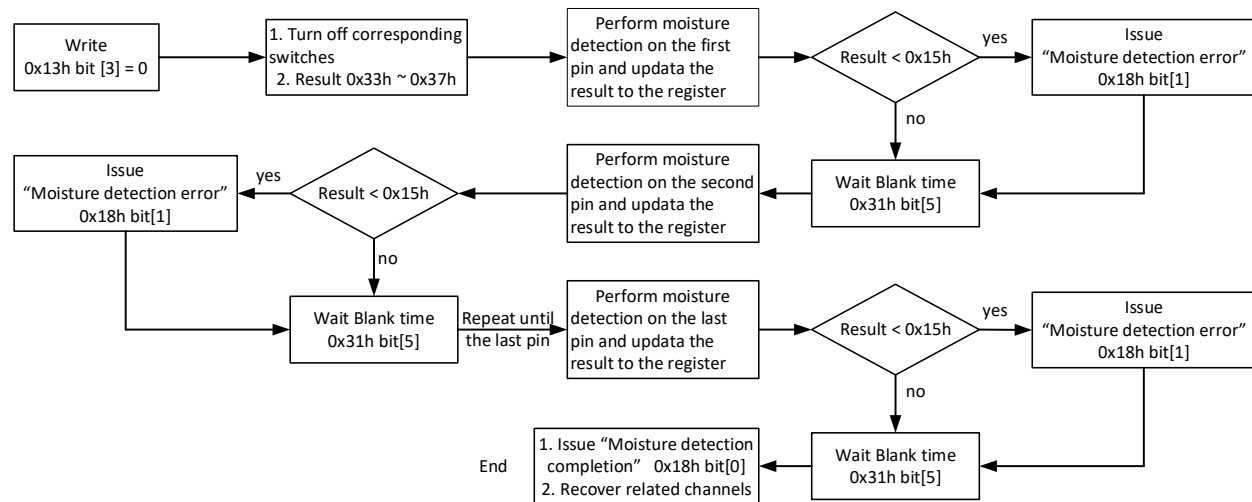


Fig. 8. Multi-pin mode (with current source enabled)

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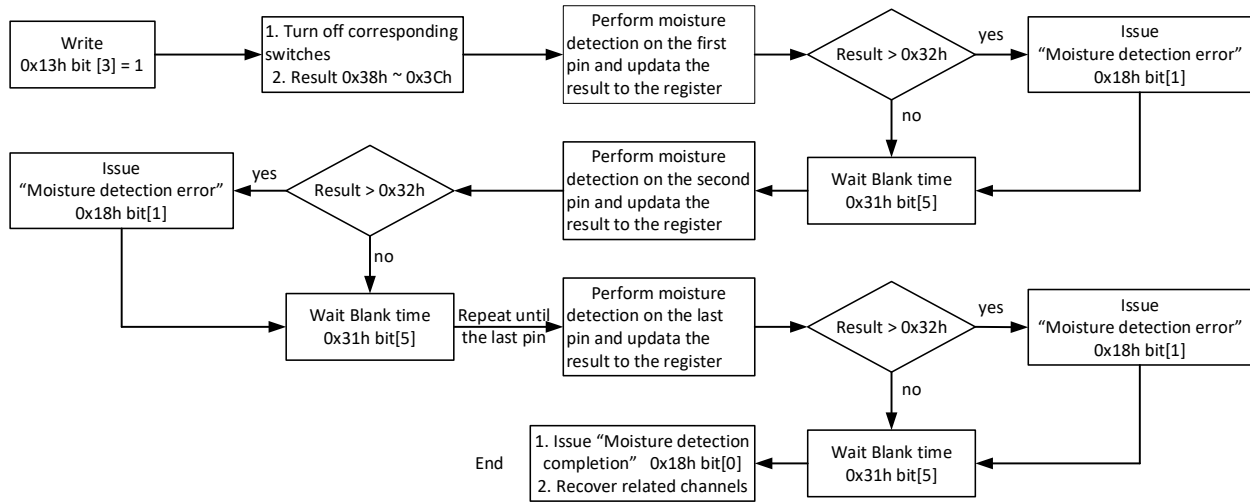


Fig. 9. Multi-pin mode (with current source disabled)

Polling

With 0x16h bit [1:0] = 1, EMS4485 can also repeatedly perform moisture detection on its own. A list of scenarios with the combination set of 0x16h bit [1:0] is shown as below. In single-pin mode, when there's no moisture detected (result ≤ 0x32h for detection with current source disabled, or result ≥ 0x15h with current source enabled), the moisture detection will be restarted after a period set by 0x16h bit [1:0].

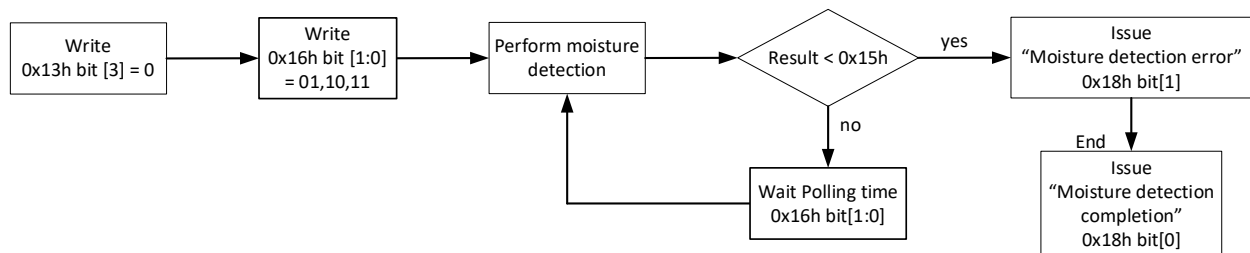


Fig. 10. Single-pin polling mode (with current source enabled)

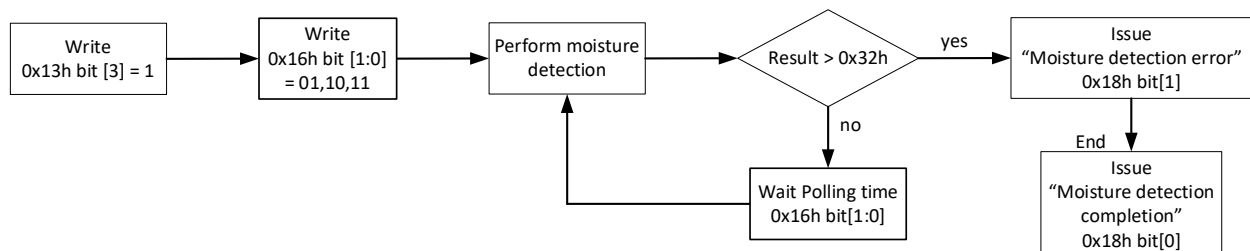


Fig. 11. Single-pin polling mode (with current source disabled)

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In multi-pin mode, when the measured result is ok after one round detection for selected pins, a new round will restart after the period set by Polling time. In the polling period, any attempt of changing the setup of current source or selecting the pins will be noticed in preparation of the next new round.

During the detection, all the related channels will be closed as soon as the first pin detection starts and the channels will not recover until the last pin detection finishes, which means this round detection is completed. To exit polling immediately, there are two situations: one situation is writing 0x13h bit [7] = 1, the other one is an OVP event detected.

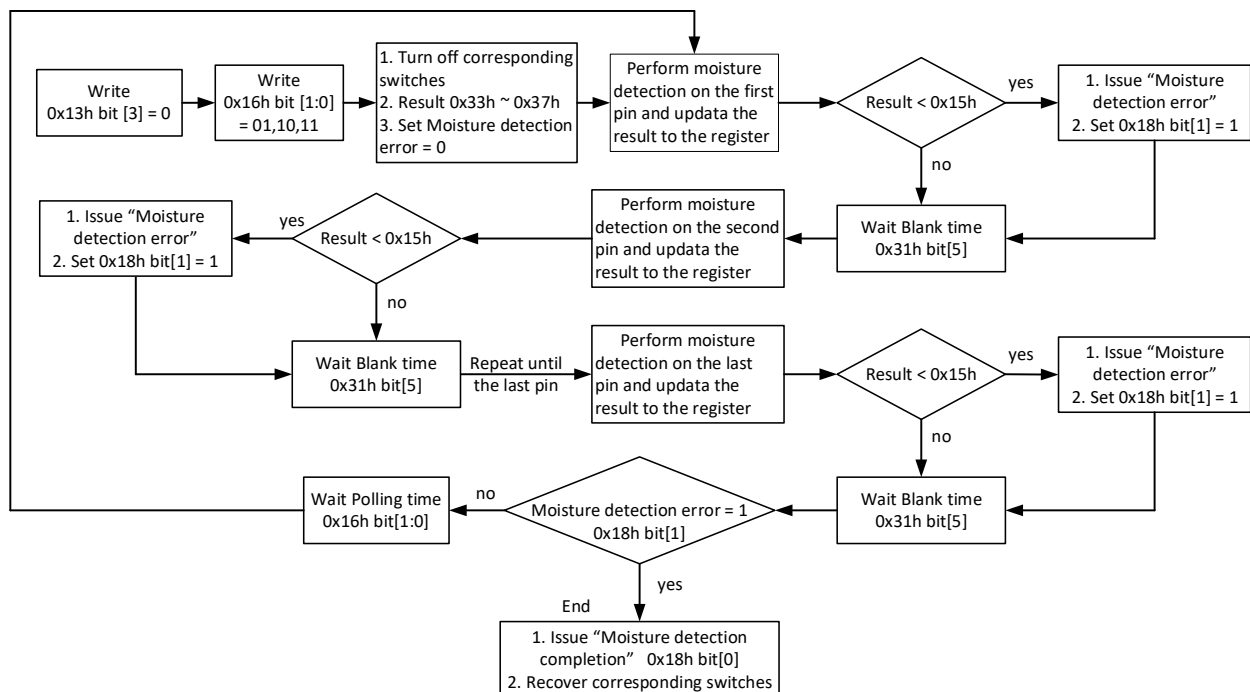


Fig. 12. Multi-pin polling mode (with current source enabled)

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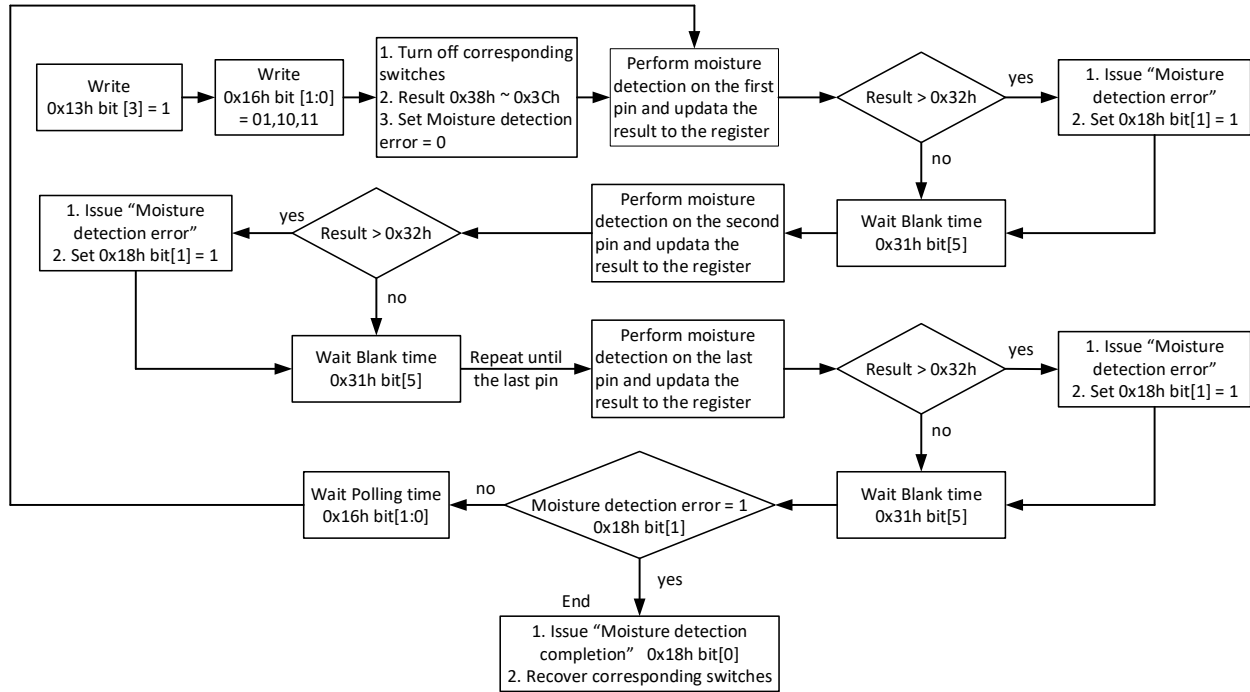


Fig. 13. Multi-pin polling mode (with current source disabled)

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12. Register Maps

ADDR	Register Name	Type	Reset Value	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
00H	Device ID	R	0xF6	1	1	1	1	0	1	0	1
02H	OVP interrupt flag	R/C	0x00	Reserved	Reserved	OVP/DP_R	OVP/DN_L	OVP/SBU1, SBU2[3:2]		OVP/GSBU1	OVP/GSBU2
03H	OVP status	R	0x00	Reserved [7:6]		OVP/DP_R	OVP/DN_L	OVP/SBU1, SBU2[3:2]		OVP/GSBU1	OVP/GSBU2
04H	Switch settings enable	R/W	0x98	Device enable	SBU1_H to SBUx switches	SBU2_H to SBUx switches	DN_L to DN or L switches	DP_R to DP or R switches	Sense to GSBUx switches	MIC to SBUx switches	AGND to SBUx switches
05H	Switch select	R/W	0x18	USB2 switch select	SBU1_H switches	SBU2_H switches	DN_L to DN1 or L switches	DP_R to DP1 or R switches	Sense to GSBUx switches	MIC to SBUx switches	AGND to SBUx switches
06H	Switch status0	R	0x05	Reserved [7:6]		Sense switch status [5:4]		DP_R switch status [3:2]		DN_L switch status [1:0]	
07H	Switch status1	R	0x00	Reserved [7:6]		SBU2 switch status [5:3]			SBU1 switch status [2:0]		
08H	Audio switch left channel turn-on control	R/W	0x01	Audio switch left channel slow control							
09H	Audio switch right channel turn-on control	R/W	0x01	Audio switch right channel slow control							
0AH	MIC switch turn-on control	R/W	0x01	MIC switch slow control							
0BH	Sense switch turn-on control	R/W	0x01	Sense switch slow control							
0CH	Audio ground switch turn-on control	R/W	0x01	Audio ground switch slow control							
11H	Audio accessory status	R	0x02	Reserved [7:2]						CC_IN	Reserved
12H	Function enable	R/W	0x08	OVP threshold voltage configuration [7:6]		Current source for resistance detection	Role of Pin C2 and C3	Slow turn on control enable	MIC auto break out control enable	Moisture detection enable	Audio jack detection & configuration enable
13H	Detection control0	R/W	0x00	Moisture detection force off control	Reserved [6:4]			Current source for moisture detection	Select the pin for moisture detection in single-pin mode [2:0]		
14H	Single-pin mode resistor detection results	R	0xFF	Moisture detection result in single-pin mode with current source enabled							
15H	Moisture resistor detection threshold	R/W	0x16	Threshold moisture detection with current source enabled							
16H	Function control1	R/W	0x00	Reserved [7:3]					Select moisture detection mode	Set the time span between two moisture detections [1:0]	
17H	MIC detection status	R	0x01	Reserved [7:4]				4 pole	4 pole	3 pole	No audio

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18H	System flag	R/C	0x00	Reserved [7:4]			Watchdog timeout	Audio jack detection & configuration	Moisture detection error	Moisture detection completion	
19H	System mask	R/W	0x00	Reserved [7:4]			Watchdog timeout mask	Audio jack detection & configuration mask	Moisture detection error mask	Moisture detection completion mask	
1CH	MIC detection threshold data0	R/W	0x20	MIC detection threshold data							
1EH	I2C reset	W/C	0x00	Reserved [7:1]							I2C reset
1FH	Bias current setting	R/W	0x07	Reserved [7:3]				Bias current setting [2:0]			
20H	Watchdog setting	R/W	0x03	Watchdog enable	Reserved [6:3]			Watchdog timer [2:0]			
30H	Detection data1	R	0x00	Moisture detection result in single-pin mode with current source disabled							
31H	Detection control1	R/W	0x00	Reserved [7:6]		Multi-pin mode moisture detection blank time	Select pin (CC_IN)	Select pin (DP_R)	Select pin (DN_L)	Select pin (SBU1) Select pin (SBU2)	
32H	Moisture voltage detection threshold	R/W	0x00	Threshold for moisture detection with current source disabled							
33H	Multi-pin mode resistor detection result of CC_IN	R	0xFF	Moisture detection result in multi-pin mode with current source enabled							
34H	Multi-pin mode resistor detection result of DP_R	R	0xFF	Moisture detection result in multi-pin mode with current source enabled							
35H	Multi-pin mode resistor detection result of DN_L	R	0xFF	Moisture detection result in multi-pin mode with current source enabled							
36H	Multi-pin mode resistor detection result of SBU1	R	0xFF	Moisture detection result in multi-pin mode with current source enabled							
37H	Multi-pin mode resistor detection result of SBU2	R	0xFF	Moisture detection result in multi-pin mode with current source enabled							
38H	Multi-pin mode voltage detection result of CC_IN	R	0x00	Moisture detection result in multi-pin mode with current source disabled							
39H	Multi-pin mode voltage detection result of DP_R	R	0x00	Moisture detection result in multi-pin mode with current source disabled							
3AH	Multi-pin mode voltage detection result of DN_L	R	0x00	Moisture detection result in multi-pin mode with current source disabled							
3BH	Multi-pin mode voltage detection result of SBU1	R	0x00	Moisture detection result in multi-pin mode with current source disabled							
3CH	Multi-pin mode voltage detection result of SBU2	R	0x00	Moisture detection result in multi-pin mode with current source disabled							

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12.1. I²C slave address

Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
1	0	0	0	0	1	0	R/W

12.2. Register definition

12.2.1. Device ID

Address: 00h

Reset Value: 8'b 1111_0110

Type: Read

Bits	Name	Size	Description
[7:6]	Vendor ID	2	Vendor ID
[5:3]	Version ID	3	Device version ID
[2:0]	Revision ID	3	Revision history ID

12.2.2. OVP interrupt flag

Address: 02h

Reset Value: 8'b 0000_0000

Type: Read Clear

Bits	Name	Size	Description
[7:6]	Reserved	2	Do not use
5	DP_R OVP	1	0: OVP event has not occurred 1: OVP event has occurred
4	DN_L OVP	1	0: OVP event has not occurred 1: OVP event has occurred
[3:2]	SBU1 SBU2 OVP	2	0: OVP event has not occurred 1: At least one of SBU1 or SBU2 OVP event has occurred
1	GSRU1 OVP	1	0: OVP event has not occurred 1: OVP event has occurred
0	GSRU2 OVP	1	0: OVP event has not occurred 1: OVP event has occurred

12.2.3. OVP status

Address: 03h

Reset Value: 8'b 0000_0000

Type: Read

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Bits	Name	Size	Description
[7:6]	Reserved	2	Do not use
5	OVP on DP_R PIN	1	0: OVP event has not occurred 1: OVP event has occurred
4	OVP on DN_L PIN	1	0: OVP event has not occurred 1: OVP event has occurred
[3:2]	SBU1 SBU2 OVP	2	0: OVP event has not occurred 1: At least one of SBU1 or SBU2 OVP event has occurred
1	OVP on GSBUS1 PIN	1	0: OVP event has not occurred 1: OVP event has occurred
0	OVP on GSBUS2 PIN	1	0: OVP event has not occurred 1: OVP event has occurred

12.2.4. Switching setting enable

Address: 04h

Reset Value: 8'b 1001_1000

Type: Read/Write

Bits	Name	Size	Description
7	Device enable	1	1: Device enable. 0: Device disable; L, R pull down by 10 kΩ and other switch nodes will be high - Z for positive input.
6	SBU1_H to SBUx switches	1	0: Switch disable; SBU1_H will be pull-down to ground. 1: Switch enable
5	SBU2_H to SBUx switches	1	0: Switch disable; SBU2_H will be pull-down to ground. 1: Switch enable
4	DN_L to DN1/2 or L switches	1	0: Switch disable; DN_L, DN1/2 will be high - Z for positive input. L pull down by 10 kΩ. 1: Switch enable
3	DP_R to DP1/2 or R switches	1	0: Switch disable; DP_R, DP1/2 will be high - Z for positive input. R pull down by 10 kΩ. 1: Switch enable
2	Sense to GSBUSx switches	1	0: Switch disable; Sense, GSBUS1 and GSBUS2 will be pull-down to ground. 1: Switch enable
1	MIC to SBUx switches	1	0: Switch disable; MIC will be pull-down to ground. 1: Switch enable
0	AGND to SBUx switches	1	0: Switch disable; AGND will be high - Z for positive input. 1: Switch enable

12.2.5. Switch select

Address: 05h

Reset Value: 8'b 0001_1000

Type: Read/Write

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Bits	Name	Size	Description
7	USB2 switch select	1	0: USB2 off. USB1 or Audio switch depend on 05h, bit [4:3] and 04h, bit [4:3] 1: DP/R~DP2, DN/L~DN2 switches ON, if 04h, bit [4:3] = '11', and 05h, bit [4:3] = '11'
6	SBU1_H switches	1	0: SBU1_H to SBU1 switch ON 1: SBU1_H to SBU2 switch ON
5	SBU2_H switches	1	0: SBU2_H to SBU2 switch ON 1: SBU2_H to SBU1 switch ON
4	DN_L to DN1 or L switches	1	0: DN_L to L switch ON 1: DN_L to DN1 switch ON
3	DP_R to DP1 or R switches	1	0: DP_R to R switch ON 1: DP_R to DP1 switch ON
2	Sense to GSBUX switches	1	0: Sense to GSBU1 switch ON 1: Sense to GSBU2 switch ON
1	MIC to SBUX switches	1	0: MIC to GSBU2 switch ON 1: MIC to GSBU1 switch ON
0	AGND to SBUX switches	1	0: AGND to SBU1 switch ON 1: AGND to SBU2 switch ON

Note: If want to change to USB2 mode, you must first write 0x05h = 0x98h, then write 0x04h = 0x98h

12.2.6. Switch status0

Address: 06h

Reset Value: 8'b 0000_0101

Type: Read

Bits	Name	Size	Description
[7:6]	Reserved	2	Do not use
[5:4]	Sense switch status	2	00: Sense switch is Open / Not Connected 01: Sense connected to GSBU1 10: Sense connected to GSBU2 11: Not valid
[3:2]	DP_R switch status	2	00: DP_R Switch Open / Not Connected 01: DP_R connected to DP1 10: DP_R connected to R 11: DP_R connected to DP2
[1:0]	DN_L switch status	2	00: DN_L Switch Open / Not Connected 01: DN_L connected to DN1 10: DN_L connected to L 11: DN_L connected to DN2

12.2.7. Switch status1

Address: 07h

Reset Value: 8'b 0000_0000

Type: Read

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Bits	Name	Size	Description
[7:6]	Reserved	2	Do not use
[5:3]	SBU2/GSBU2 switch status	3	000: SBU2/GSBU2 switch is Open / Not Connected 001: SBU2/GSBU2 connected to MIC 010: SBU2/GSBU2 connected to AGND 011: SBU2/GSBU2 connected to SBU1_H 100: SBU2/GSBU2 connected to SBU2_H 101: SBU2/GSBU2 connected both SBU1_H and SBU2_H 110...111: Do not use
[2:0]	SBU1/GSBU1 switch status	3	000: SBU1/GSBU1 switch is Open / Not Connected 001: SBU1/GSBU1 connected to MIC 010: SBU1/GSBU1 connected to AGND 011: SBU1/GSBU1 connected to SBU1_H 100: SBU1/GSBU1 connected to SBU2_H 101: SBU1/GSBU1 connected both SBU1_H and SBU2_H 110...111: Do not use

12.2.8. Audio switch left channel slow turn-on

Address: 08h

Reset Value: 8'b 0000_0001

Type: Read/Write

Bits	Name	Size	Description
[7:0]	Switch turn on rising time setting	8	11111111: 8300 μ s ... 10111111: 645 μ s ... 01111111: 327 μ s ... 00111111: 222 μ s ... 00000001: 114 μ s

12.2.9. Audio switch right channel slow turn-on

Address: 09h

Reset Value: 8'b 0000_0001

Type: Read/Write

Bits	Name	Size	Description
[7:0]	Switch turn on rising time setting	8	11111111: 8300 μ s ... 10111111: 645 μ s ... 01111111: 327 μ s ... 00111111: 222 μ s ... 00000001: 114 μ s

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12.2.10. MIC switch slow turn-on

Address: 0Ah

Reset Value: 8'b 0000_0001

Type: Read/Write

Bits	Name	Size	Description
[7:0]	Switch turn on rising time setting	8	11111111: 57567 μ s ... 10111111: 7680 μ s ... 01111111: 4833 μ s ... 00111111: 3883 μ s ... 00000001: 2067 μ s

12.2.11. Sense switch slow turn-on

Address: 0Bh

Reset Value: 8'b 0000_0001

Type: Read/Write

Bits	Name	Size	Description
[7:0]	Switch turn-on rising time setting	8	11111111: 166000 μ s ... 10111111: 10883 μ s ... 01111111: 5017 μ s ... 00111111: 3303 μ s ... 00000001: 1273 μ s

12.2.12. Audio ground switch slow turn-on

Address: 0Ch

Reset Value: 8'b 0000_0001

Type: Read/Write

Bits	Name	Size	Description
[7:0]	Switch turn-on rising time setting	8	11111111: 17500 μ s ... 10111111: 1327 μ s ... 01111111: 632 μ s ... 00111111: 427 μ s ... 00000001: 263 μ s

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12.2.13. Audio accessory status

Address: 11h

Reset Value: 8'b 0000_0010

Type: Read

Bits	Name	Size	Description
[7:2]	Reserved	6	Do not use
1	CC_IN	1	0: CC_IN < 1.2 V 1: CC_IN > 1.5 V
0	Reserved	1	Do not use

12.2.14. Function enable

Address: 12h

Reset Value: 8'b 0000_1000

Type: Read/Write

Bits	Name	Size	Description
[7:6]	OVP threshold voltage configuration	2	00: 4.4 V 01: 4.6 V 10: 4.8 V 11: 5.0 V
5	Current source for resistance detection	1	If 0× 13h bit [3] = 0. 0: 5 μ A 1: 1 μ A
4	Role of pin C2 and C3	1	0: Set C2 as DET and C3 as INT 1: Set C2 as COMPIN and C3 as COMPOUT
3	Slow turn on control enable	1	1: Enable 0: Disable
2	MIC auto break out control enable	1	1: Enable 0: Disable
1	Moisture detection enable	1	1: Enable. It would return to '0' after the detection was finished. 0: Disable (This bit can be reset by POR, I ² C or WTD)
0	Audio jack detection and configuration enable	1	1: Enable. It will be changed to '0' after audio jack detection and configuration 0: Disable

12.2.15. Detection control0

Address: 13h

Reset Value: 8'b 0000_0000

Type: Read/Write

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Bits	Name	Size	Description
7	Moisture detection force off control	1	0: Moisture detection on and off control would follow 0×12h bit [1] 1: Moisture detection would be turned off forcibly
[6:4]	Reserved	3	Do not use
3	Current source for moisture detection	1	0: Enable 1: Disable
[2:0]	Select the pin for moisture detection in single-pin mode	3	000: CC_IN 001: DP_R 010: DN_L 011: SBU1 100: SBU2 101...111: Reserved

12.2.16. Single-pin mode resistor detection results

Address: 14h

Reset Value: 8'b 1111_1111

Type: Read

Bits	Name	Size	Description
[7:0]	Moisture detection result in single-pin mode with current source enabled	8	0x00 ~ 0xFF: 2 kΩ * Bits, with 0×12h bit [5] = 0; 10 kΩ * Bits, with 0×12h bit [5] = 1.

12.2.17. Moisture resistor detection threshold

Address: 15h

Reset Value: 8'b 0001_0110

Type: Read/Write

Bits	Name	Size	Description
[7:0]	Threshold for moisture detection with current source enabled.	8	Would issue 'Moisture detection error' once any contents from 0×14h (in single pin mode) or 0×33h ~ 0×37h (in multi-pin mode) is lower than this threshold.

12.2.18. Function control1

Address: 16h

Reset Value: 8'b 0000_0000

Type: Read/Write

Bits	Name	Size	Description
[7:3]	Reserved	5	Do not use.
2	Select moisture detection mode	1	0: Single-pin mode, will perform moisture on single pin. 1: Multi-pin mode, will perform moisture on multiple pins.

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[1:0]	Set the time span between two moisture detections	2	00: No polling. In single-pin mode, perform single detection for the selected pin (0×13h bit [2:0]). In multi-pin mode, perform single detection for all selected pins (0×31h bit [4:0]). 01: Polling with 400 ms time span 10: Polling with 2s time span 11: Polling with 20s time span
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12.2.19. MIC detection status

Address: 17h

Reset Value: 8'b 0000_0001

Type: Read

Bits	Name	Size	Description
[7:4]	Reserved	4	Do not use
3	4 pole	1	1: 4 Pole SBU2 to MIC, SBU1 to audio ground 0: others
2	4 pole	1	1: 4 Pole SBU1 to MIC, SBU2 to audio ground 0: others
1	3 pole	1	1: 3 pole 0: others
0	No audio accessory	1	1: No audio accessory 0: Audio accessory attached

12.2.20. System flag

Address: 18h

Reset Value: 8'b 0000_0000

Type: Read Clear

Bits	Name	Size	Description
[7:4]	Reserved	4	Do not use
3	Watchdog timeout	1	0: Watchdog timeout has not occurred 1: Watchdog timeout has occurred
2	Audio jack detection and configuration	1	0: Audio jack detection and configuration have not completed. 1: Audio jack detection and configuration is completed.
1	Moisture detection error	1	0: Moisture doesn't occur. For resistance detection, result(s) $\geq$ TH0 For voltage detection, result(s) $\leq$ TH1 1: Moisture occurs. For resistance detection, result(s) $<$ TH0 For voltage detection, result(s) $>$ TH1
0	Moisture detection completion	1	0: Moisture detection has been not completed 1: Moisture detection is completed.

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12.2.21. System mask

Address: 19h

Reset Value: 8'b 0000_0000

Type: Read/Write

Bits	Name	Size	Description
[7:4]	Reserved	4	Do not use.
3	Watchdog timeout mask	1	0: "Watchdog timeout" bit is not masked. 1: "Watchdog timeout" bit is masked.
2	Audio jack detection and configuration mask	1	0: "Audio jack detection and configuration" bit is not masked. 1: "Audio jack detection and configuration" bit is masked.
1	Moisture detection error mask	1	0: "Moisture detection error" bit is not masked. 1: "Moisture detection error" bit is masked.
0	Moisture detection completion mask	1	0: "Moisture detection completion" bit is not masked. 1: "Moisture detection completion" bit is masked.

12.2.22. MIC detection threshold data

Address: 1Ch

Reset Value: 8'b 0010_0000

Type: Write

Bits	Name	Size	Description
[7:0]	MIC detection threshold data	8	MIC detection threshold data.

12.2.23. I²C reset

Address: 1Eh

Reset Value: 8'b 0000_0000

Type: Write / Clear

Bits	Name	Size	Description
[7:1]	Reserved	7	Reserved
0	I ² C reset	1	0: default 1: I ² C reset

12.2.24. Bias current setting

Address: 1Fh

Reset Value: 8'b 0000_0111

Type: Read/Write

Bits	Name	Size	Description
[7:3]	Reserved	4	Do not use

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[2:0]	Bias current setting	4	Setting bias current for audio jack detection. 0000 ~ 1111: Bits × 100 μ A
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12.2.25. Watchdog setting

Address: 20h

Reset Value: 8'b 0000_0011

Type: Read/Write

Bits	Name	Size	Description
7	Watchdog enable	1	0: Watchdog disabled 1: Watchdog enabled
[6:3]	Reserved	4	Do not use
[2:0]	Watchdog timer	3	000: 0.5 s 001: 1 s 010: 2 s 011: 5 s 100: 10 s 101: 30 s 110: 60 s 111: 5 min

12.2.26. Detection data1

Address: 30h

Reset Value: 8'b 0000_0000

Type: Read

Bits	Name	Size	Description
[7:0]	Detection data1	8	Moisture detection result in single-pin mode with current source disabled: 0x00 ~ 0xFF: Bits × 9 × 0.001 [Unit: V]

12.2.27. Detection control1

Address: 31h

Reset Value: 8'b 0000_0000

Type: Read/Write

Bits	Name	Size	Description
[7:6]	Reserved	2	Do not use.
5	Multi-pin mode moisture detection blank time	1	In multi-pin mode, set blank time between detections for different pins. 0: 20 ms 1: 120 ms
4	Select pin (CC_IN)	1	Select pin for moisture detection in multi-pin mode. 0: not selected 1: selected
3	Select pin (DP_R)	1	Select pin for moisture detection in multi-pin mode. 0: not selected 1: selected

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2	Select pin (DN_L)	1	Select pin for moisture detection in multi-pin mode. 0: not selected 1: selected
1	Select pin (SBU1)	1	Select pin for moisture detection in multi-pin mode. 0: not selected 1: selected
0	Select pin (SBU2)	1	Select pin for moisture detection in multi-pin mode. 0: not selected 1: selected

12.2.28. Moisture voltage detection threshold

Address: 32h

Reset Value: 8'b 0000_0000

Type: Read/Write

Bits	Name	Size	Description
[7:0]	Threshold for moisture detection with current source disabled	8	Threshold for moisture detection with current source disabled. EMS4485 would issue "Moisture detection error" as soon as any contents from 0x30h (in single-pin mode) or 0x38h ~ 0x3Ch (in multi-pin mode) is higher than this threshold.

12.2.29. Multi-pin mode resistor detection result of CC_IN

Address: 33h

Reset Value: 8'b 1111_1111

Type: Read

Bits	Name	Size	Description
[7:0]	Moisture detection in multi-pin mode (CC_IN)	8	Moisture detection result in multi-pin mode with current source enabled: 0x00 ~ 0xFF: 2 k Ω * Bits, with 0x 12h bit [5] = 0; 10 k Ω * Bits, with 0x 12h bit [5] = 1.

12.2.30. Multi-pin mode resistor detection result of DP_R

Address: 34h

Reset Value: 8'b 1111_1111

Type: Read

Bits	Name	Size	Description
[7:0]	Moisture detection in multi-pin mode (DP_R)	8	Moisture detection result in multi-pin mode with current source enabled: 0x00 ~ 0xFF: 2 k Ω * Bits, with 0x 12h bit [5] = 0; 10 k Ω * Bits, with 0x 12h bit [5] = 1.

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12.2.31. Multi-pin mode resistor detection result of DN_L

Address: 35h

Reset Value: 8'b 1111_1111

Type: Read

Bits	Name	Size	Description
[7:0]	Moisture detection in multi-pin mode (DN_L)	8	Moisture detection result in multi-pin mode with current source enabled: 0x00 ~ 0xFF: 2 k Ω * Bits, with 0x 12h bit [5] = 0; 10 k Ω * Bits, with 0x 12h bit [5] = 1.

12.2.32. Multi-pin mode resistor detection result of SBU1

Address: 36h

Reset Value: 8'b 1111_1111

Type: Read

Bits	Name	Size	Description
[7:0]	Moisture detection in multi-pin mode (SBU1)	8	Moisture detection result in multi-pin mode with current source enabled: 0x00 ~ 0xFF: 2 k Ω * Bits, with 0x 12h bit [5] = 0; 10 k Ω * Bits, with 0x 12h bit [5] = 1.

12.2.33. Multi-pin mode resistor detection result of SBU2

Address: 37h

Reset Value: 8'b 1111_1111

Type: Read

Bits	Name	Size	Description
[7:0]	Moisture detection in multi-pin mode (SBU2)	8	Moisture detection result in multi-pin mode with current source enabled: 0x00 ~ 0xFF: 2 k Ω * Bits, with 0x 12h bit [5] = 0; 10 k Ω * Bits, with 0x 12h bit [5] = 1.

12.2.34. Multi-pin mode voltage detection result of CC_IN

Address: 38h

Reset Value: 8'b 0000_0000

Type: Read

Bits	Name	Size	Description
[7:0]	Moisture detection in multi-pin mode (CC_IN)	8	Moisture detection result in multi-pin mode with current source disabled. 0x00 ~ 0xFF: Bits $\times$ 9 $\times$ 0.001 [unit: V]

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Address: 39h

Reset Value: 8'b 0000_0000

Type: Read

Bits	Name	Size	Description
[7:0]	Moisture detection in multi-pin mode (DP_R)	8	Moisture detection result in multi-pin mode with current source disabled. 0x00 ~ 0xFF: Bits $\times 9 \times 0.001$ [unit: V]

12.2.36. Multi-pin mode voltage detection result of DN_L

Address: 3Ah

Reset Value: 8'b 0000_0000

Type: Read

Bits	Name	Size	Description
[7:0]	Moisture detection in multi-pin mode (DN_L)	8	Moisture detection result in multi-pin mode with current source disabled. 0x00 ~ 0xFF: Bits $\times 9 \times 0.001$ [unit: V]

12.2.37. Multi-pin mode voltage detection result of SBU1

Address: 3Bh

Reset Value: 8'b 0000_0000

Type: Read

Bits	Name	Size	Description
[7:0]	Moisture detection in multi-pin mode (SBU1)	8	Moisture detection result in multi-pin mode with current source disabled. 0x00 ~ 0xFF: Bits $\times 9 \times 0.001$ [unit: V]

12.2.38. Multi-pin mode voltage detection result of SBU2

Address: 3Ch

Reset Value: 8'b 0000_0000

Type: Read

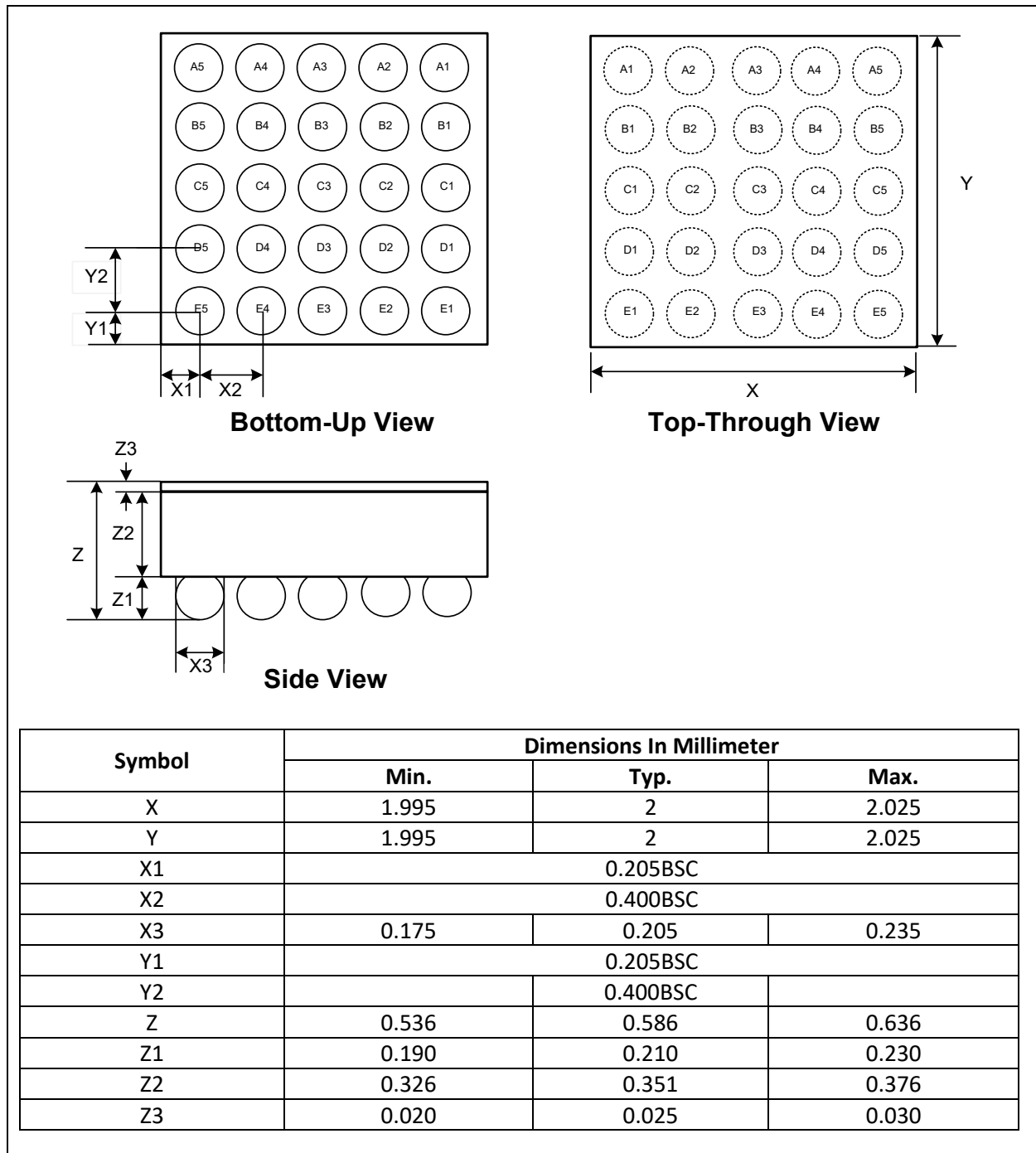
Bits	Name	Size	Description
[7:0]	Moisture detection in multi-pin mode (SBU2)	8	Moisture detection result in multi-pin mode with current source disabled. 0x00 ~ 0xFF: Bits $\times 9 \times 0.001$ [unit: V]

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13. Package Outline

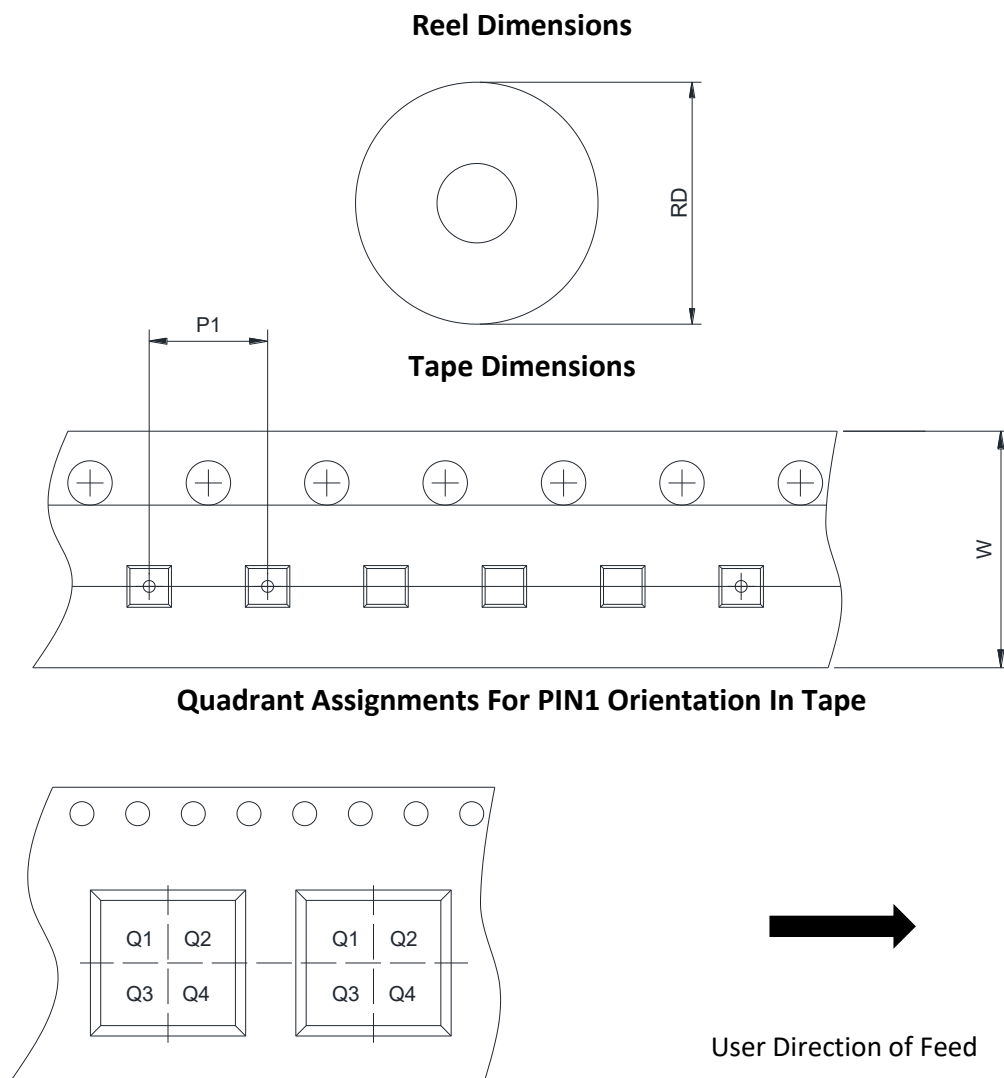
WLCSP-25



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14. Tape and Reel Information



RD	Reel Dimension	☒ 7inch ☐ 13inch
W	Overall width of the carrier tape	☒ 8mm ☐ 12mm ☐ 16mm
P1	Pitch between successive cavity centers	☐ 2mm ☒ 4mm ☐ 8mm
Pin1	Pin1 Quadrant	☒ Q1 ☐ Q2 ☐ Q3 ☐ Q4

15. Revision History

Table 8. Revision history

Document ID	Release Date	Data sheet status	Change notice	Supersedes
EMS4485 Rev. 0.9	Jul 06, 2026	Draft datasheet		EMS4485 Rev. 0.1
Modifications:	- Section 3: Changed the package name from WLCSP-25B to WLCSP-25 - Section 4: Added pin connection notes for AGND/GND, SBU1/GSBU1, and SBU2/GSBU2. - Section 7: Updated Absolute Maximum Ratings: V_{SW_C}, V_{SW_Audio}, V_{SW_USB}, V_{SW_UART}, V_{VSBUX_H}, $V_{I/O}$, V_{CNTRL}, $V_{comparator}$, I_{SW_UART} updated. - Section 8: Updated Recommended Operating Conditions: V_{CC}, V_{SW_Audio}, V_{SW_UART}, V_{VSBUX}, V_{IH} updated. - Section 9: Updated ESD Ratings - Section 10: Updated DC electrical characteristics (Test condition V_{CC} max changed from 5.5V to 4.5V): I_{ON} (Audio), I_{ON} (UART) updated. R_{FLAT} removed. R_{OFF} (Audio) added. - Section 10: Updated AC electrical characteristics (Test condition V_{CC} max changed from 5.5V to 4.5V): Audio switch t_{delay}, Audio switch t_{rise}, USB switch t_{ON}/t_{OFF}, MIC/AGND switch timing updated. UART switch t_{ON}/t_{OFF} added. - Section 10: Added 10.3 Comparator electrical characteristics and 10.4 Comparator electrical characteristics - Section 12: Updated Register Maps 0x04h: Disabled state behavior changed from high-Z to pull-down to ground (Bits 6, 5, 2, 1). 0x07h: Register names updated to SBU2/GSBU2 status (Bits [5:3]) and SBU1/GSBU1 status (Bits [2:0]). 0x08h~0x0Ch: R/L Switch, MIC switch, Sense switch, Audio ground switch turn-on rising time ranges updated. 0x16h: Polling time span options updated. 0x31h: Multi-pin mode blank time options updated.			
EMS4485 Rev. 0.1	May 06, 2026	Draft datasheet		